

Problem

A series RLC circuit has $R = 10\ \Omega$, $L = 2\text{ mH}$, and $C = 40\ \mu\text{F}$. Determine the effective current and average power absorbed when the applied voltage is

$$v(t) = 100 \cos 1000t + 50 \cos 2000t + 25 \cos 3000t\text{ V}$$

Step-by-step solution

Step 1 of 8

The applied voltage is,

$$v(t) = 100 \cos 1000t + 50 \cos 2000t + 25 \cos 3000t\text{ V}$$

Consider $100 \cos 1000t$ source.

Calculate the impedance value of inductor.

$$\begin{aligned} X_L &= \omega L \\ &= (1000)(2 \times 10^{-3}) \\ &= 2\ \Omega \end{aligned}$$

Calculate the impedance value of capacitor.

$$\begin{aligned} X_C &= \frac{1}{\omega C} \\ &= \frac{1}{(1000)(40 \times 10^{-6})} \\ &= 25\ \Omega \end{aligned}$$

Step 2 of 8

Calculate the total impedance.

$$\begin{aligned} Z &= R + jX_L - jX_C \\ &= 10 + j2 - j25 \\ &= 10 - j23\ \Omega \end{aligned}$$

Calculate the rms current.

$$\begin{aligned} I_1 &= \frac{100}{10 - j23} \\ &= 1.589 + j3.6566 \\ &= 3.987 \angle 66.5^\circ\text{ A} \end{aligned}$$

Step 3 of 8

Consider $50 \cos 2000t$ source.

Calculate the impedance value of inductor.

$$\begin{aligned}X_L &= \omega L \\&= (2000)(2 \times 10^{-3}) \\&= 4 \, \Omega\end{aligned}$$

Calculate the impedance value of capacitor.

$$\begin{aligned}X_C &= \frac{1}{\omega C} \\&= \frac{1}{(4000)(40 \times 10^{-6})} \\&= 12.5 \, \Omega\end{aligned}$$

Step 4 of 8

Calculate the total impedance.

$$\begin{aligned}Z &= R + jX_L - jX_C \\&= 10 + j4 - j12.5 \\&= 10 - j8.5 \, \Omega\end{aligned}$$

Calculate the rms current.

$$\begin{aligned}I_2 &= \frac{50}{10 - j8.5} \\&= 2.902 + j2.467 \\&= 3.8097 \angle 40.36^\circ \text{ A}\end{aligned}$$

Step 5 of 8

Consider $25 \cos 3000t$ source.

Calculate the impedance value of inductor.

$$\begin{aligned}X_L &= \omega L \\&= (3000)(2 \times 10^{-3}) \\&= 6 \, \Omega\end{aligned}$$

Calculate the impedance value of capacitor.

$$\begin{aligned}X_C &= \frac{1}{\omega C} \\&= \frac{1}{(3000)(40 \times 10^{-6})} \\&= 8.333 \, \Omega\end{aligned}$$

Step 6 of 8

Calculate the total impedance.

$$\begin{aligned} Z &= R + jX_L - jX_C \\ &= 10 + j6 - j8.333 \\ &= 10 - j2.333 \, \Omega \end{aligned}$$

Calculate the rms current.

$$\begin{aligned} I_3 &= \frac{25}{10 - j2.333} \\ &= 2.371 + j0.553 \\ &= 2.435 \angle 13.13^\circ \text{ A} \end{aligned}$$

Step 7 of 8

Calculate the effective or rms current.

$$\begin{aligned} I_{\text{rms}} &= \sqrt{\frac{I_1^2 + I_2^2 + I_3^2}{2}} \\ &= \sqrt{\frac{(3.987)^2 + (3.8097)^2 + (2.435)^2}{2}} \\ &= \sqrt{\frac{36.34}{2}} \\ &= \sqrt{18.169} \\ &= 4.263 \text{ A} \end{aligned}$$

Therefore, the value of effective current is **4.263 A**.

Step 8 of 8

Calculate the average power absorbed.

$$P = I_{\text{rms}}^2 R$$

Substitute 4.263 A for I_{rms} and 10 Ω for R .

$$\begin{aligned} P &= (4.263)^2 (10) \\ &= (18.169)(10) \\ &= 181.69 \text{ W} \end{aligned}$$

Therefore, the average power absorbed is **181.69 W**.